

Method for creating a tissue microarray-capable, slide-scanning acquisition device from any motorized microscope

Steven P. Nilsen,¹ M. Lora D. L. M. Ong,¹ Jeremy Muhlich,² Jay Copeland,³ Neal Gliksman,⁴ Jerrold R. Turner¹

¹ Laboratory of Mucosal Barrier Pathobiology, Department of Pathology, Brigham and Women's Hospital and Harvard Medical School, Boston, MA 02115

² Department of Systems Biology, Harvard Medical School, Boston, MA 02115

³ Research Information Technology Solutions, Harvard Medical School, Boston, MA 02115

⁴ Molecular Devices, West Chester, Pennsylvania 19380

Correspondence to:

Jerrold R. Turner, PhD, MD, Departments of Pathology and Medicine, Brigham and Women's Hospital and Harvard Medical School, Boston, MA, USA, Tel +16175258165, email: jrturner@bwh.harvard.edu ORCID: 0000-0003-0627-9455

Support:

Supported by the National Institute of Diabetes Digestive and Kidney Disease (R01DK61931, R01DK068271, R24DK099803) and the Harvard Digestive Disease Center (P30DK034854).

Abstract:

High-throughput, multiplex slide-scanning technologies have become widely-used. Advantages include the ability to archive digital copies of slides, review slides as teams using virtual microscope software, and standardize analytical approaches. One barrier to implementation is the cost of dedicated slide-scanning devices. We describe a simple method that allows any microscope to be used for slide-scanning. The only requirements are that the microscope be equipped with a motorized filter turret (for multichannel fluorescence) and a motorized xyz stage. This example uses Metamorph, the most commonly used hardware-agnostic microscope control software. The approach can, however, be readily translated to open source applications as well as to dedicated control software provided by microscope manufacturers. The scripts, or journals, developed allow users to define multiple regions of interest and specific parameters, including ranges and channels used for focusing. Following acquisition, captured images can be stitched and displayed using open source virtual microscope software.

Introduction:

As with all research, analysis of microscopic images is increasingly quantitative. These assessments rely on digital images using cameras with broad, linear dynamic ranges. However, only selected fields are generally captured. This limits the ability of other observers to evaluate the entire specimen and creates the possibility of intentional or unintentional sampling bias. One way to overcome this is to image the entire specimen. This is particularly advantageous for fluorescent samples, e.g., immunofluorescent stained cells and tissues, in which repeated viewing causes photobleaching, and other samples that degrade over time. Thus, whole-slide imaging effectively preserves the specimen such that it can be viewed repeatedly and archived without fear of damage. This approach also allows i) review on any computer without need for a microscope; ii) display of whole-slide images in group meetings, for both analysis and education of new investigators; iii) simple image annotation; iv) post-acquisition analysis, and v) sharing of unfiltered, raw data between investigators at different sites. As a result of these advantages, most software packages that control microscopes now include a slide scanning module that allows users to capture a large area region of interest (ROI) as many smaller fields (tiles). The individual images are then stitched together to create a composite that provides both low and high magnification views for use on virtual microscope platforms.

Available slide scanning modules typically allow for only a single ROI to be imaged at once. This imparts a significant limitation on the increasing use of tissue microarrays (TMAs) and other technologies that enable the staining of many samples on a single slide.^{1,2} One could avoid this limitation by imaging the entire slide, but this would greatly increase the time required for data collection and results in enormous data files that are both difficult to stitch and difficult to view. Multiple manufacturers have circumvented this challenge by developing automated slide scanning systems that only image regions of slides where specimens are present. These turnkey systems work well, but are generally only available in core facilities and cannot be customized. Moreover, images can only be reviewed using proprietary software and costly hardware. We sought to develop a similar automated slide scanning system that could easily be assembled within individual labs from a simple fluorescent microscope. This is both cost-effective, relative to purchasing a new, dedicated instrument, and also provides complete flexibility in modifying any aspect of the system, including imaging modality, objectives used, illumination sources, filters, and cameras. Moreover, files are created using standard formats that are easily displayed by open-source software, such as ImageJ.³⁻⁵

- **Development of the protocol:**

We sought to overcome limitations of available turnkey, automated, slide scanning systems and slide scanning modules within more widely-available software. We required that the solution

would i) allowing for automated scanning of multiple ROIs; ii) create files of manageable sizes using standard formats, iii) include customizable focusing routines; iv) image all fields within regions of interest, even if no sample is present; and v) produce high-quality images with relatively-short imaging times.

Metamorph (Molecular Devices), which remains the most widely-used hardware-agnostic microscope control software, was used as the starting point. This allowed us to develop control scripts, termed journals in Metamorph, that could easily be applied regardless to multiple imaging modalities and microscopes, objectives, stages, light sources, filters, and cameras produced by a wide range of vendors. The journals are annotated to simplify modification for other uses and we anticipate that they can be easily translated to other proprietary and open-source microscope control software, such as μ Manager.⁶

- **Applications of the method:**

The method can be widely applied to transmitted light and fluorescence imaging using wide-field, confocal, or other, e.g. super resolution, light microscopes. It will be particularly useful for investigators using advanced preparation techniques, e.g., immunofluorescent staining, and specimen collections, e.g., tissue microarrays.

- **Comparison with other methods:**

In order to create turnkey systems that can image a wide variety of samples, many manufacturers have had to compromise in multiple areas. As a result, the systems are unsuitable for some samples. For example, some commercial fluorescent slide scanners leave slides almost completely bleached, such that they cannot be reviewed using a standard microscope. Other systems rely on objectives with long working distances and low numerical apertures (NAs) objectives to simplify focusing. However, both the light sensitivity and resolution of these objectives is inferior to those of objectives with short working distances and high NAs. Turnkey systems also generally capture the entire slide, thereby creating files that exceed size limitations of standard image file formats, approximately 4 GB for the standard TIFF format. This is necessitated development of proprietary, nonstandard file formats and specialized viewing software. Finally, customization is difficult or impossible. For example, it may not be simple to convert a system configured to image 3 specific fluorochromes to one capable of imaging 4 different fluorochromes. In contrast, our system is built on a conventional microscope, is infinitely customizable, generates standard TIFF format images. Users can decide which

objectives are preferable for their purposes. In addition, imaging time is reduced by user selection of specific ROIs and advanced focusing methods.

- **Experimental design:**

The protocol described is primarily for multi-wavelength fluorescent wide-field imaging of TMAs. It is suitable for multiplex staining approaches, in which antibodies are stripped or fluorochromes inactivated between imaging cycles, thereby allowing analysis of 20 or more markers in individual cells.⁷⁻¹¹ Methods for preparation and staining of TMAs are discussed in detail elsewhere^{7,9} as are approaches to multiplex staining and analysis.^{9, 11, 12} While this approach is ideally suited to quantitative analysis of multiple samples on a single slide, it is important to standardize imaging parameters, such as exposure time, and include internal controls, ideally replicate samples on each slide, when comparing samples on multiple slides.

- **Limitations:**

Although it is relatively simple to build and use, this is an open, do-it-yourself system. It therefore requires greater expertise than turnkey systems. Moreover, dedicated support beyond that provided for the microscope is not available.

- **Overview of the procedure:**

Stage 1: Steps 1 - 12, creation of an overview image showing all tissue locations

The first stage of the procedure is to create a single-image overview scan of the entire slide using a stain that labels all specimens. A nuclear stain, e.g. DAPI or Hoescht 33342, is ideal. Any ubiquitous stain could be used for this step, even tissue auto-fluorescence could be sufficient, but rapid imaging of a strong nuclei stain saves time and has proven reliable.

The overview scan will be used to select ROIs for acquisition of high quality, tiled images. We use a 5X objective; a 2.5X objective was tested but the resolution proved insufficient for our ~3 mm diameter samples. We took advantage of the built-in scanning functionality of Metamorph, which provides an interface for making tiled scans of each ROI (Fig. 1A). This is highly flexible and allows for scripted functions to augment and substitute some built-in routines in order to enhance quality and reduce scan times.

Stage 2: Steps 13 - 18, selecting ROIs

This stage of the procedure requires that the ROIs are manually selected from the overview scan (Fig. 1A). Manual selection provides several advantages but does take some time. Precise definition of individual ROIs (Fig. 1B) will minimize imaging of empty spaces and reduce scanning time. In addition, this allows selection of which samples are scanned. An additional advantage of this approach is that each sample will be collected precisely with numbers defined by the user. Routines used in commercial slide scanners that image the entire slide may require less interface time but can use more scanning time. If only one ROI needs to be scanned, *e.g.* a single large tissue sample, some of the scripted functions, *i.e.*, journals, developed can be used to reduce scan time by customizing focusing frequency.

Stage 3: Steps 19 - 27, setting imaging parameters and initiating the automated scan

In this stage the imaging parameters used to collect the analytical images are determined and the scripted program is run. This will result in acquisition of individual image tiles for each ROI. Some trial and error, depending on the microscopy, sample, and stain, may allow refinement of scanning efficiency by, for example, limiting focus checks to the number required.

Stage 4: Steps 28 - 31, review scan quality

It is worthwhile to review the scans for any image problems, such as focus, exposure, or ROI selection. For example, strong fluorescent debris can occasionally interfere with focusing within a ROI. Alternatively, the focal range parameters may have been set improperly, which can cause multiple ROIs to be out of focus. In other cases the ROI may have been mispositioned such that the tissue is imaged incompletely. In these cases, the problem will need to be corrected and new tiled images collected before proceeding.

Stage 5: Steps 31 – 41, (optional) rescanning TMA slides for background subtraction and antibody multiplexing

When rescanning the same TMA slides the procedure runs more efficiently by reusing the overview scan made in Stage 1, the ROI map made in Stage 2, and several of the imaging parameters determined in Stage 3. Running the automated scan and reviewing the results (Stage 4) is the most time-consuming part of this stage.

Stage 6: Steps 42 - 43, assembling the image tiles into stitched images for analysis

To keep the scanning microscope free for access, images are transferred to a more powerful, offline computer, *e.g.* a workstation, for assembly into composite stitched images. Stitching can

be done natively in Metamorph or using other programs, such as the ImageJ Stitching plugin¹³ or Python-based, open-source ASHLAR software.^{10, 14} Each approach has advantages and disadvantages.

Materials:**EQUIPMENT.**

Microscope: the method is applicable to many different microscopes, including upright and inverted wide-field and confocal microscopes from many manufacturers. This system was developed using a Leica DM4000 fluorescent microscope. An advantage of this microscope is that it has a motorized filter turret that allows use of individual excitation and emission filters for each fluorochrome. It is possible to use a multi-bandpass dichroic and emission filter and individual excitation filters or light sources, e.g., lasers or LEDs. Our experience, however, is that these capture a greater degree of nonspecific signal from tissue autofluorescence, which is detected across a wide spectrum of excitation and emission wavelengths.

Stage: a motorized stage is required. The system described is equipped with a Ludl BioPrecision2 xy stage with mounting accessories for the Leica DM4000. The stage was selected because of its high resolution ($<0.5\text{ }\mu\text{m}$ repeatability) and linear encoders that allow precise positioning. The microscope is also equipped with a Ludl focus motor with linear encoder for z axis control. Both stage and focus motor are controlled by a Ludl MAC 6000.

Specimen holder: the system described here is equipped with a Ludl 4 position slide holder (to allow 4 TMA slides to be imaged at once). Systems with single slide holders or, alternatively, automated high-capacity slide holders, can also be used. Care must be taken to ensure that slides are placed in register in the xz plane and are aligned with the camera focal plane with z variance of less than $30\text{ }\mu\text{m}$ across the slides.

Objectives: these are infinitely customizable. This system is equipped with a Leica HC PL FLUOTAR 5x 0.15NA objective for preliminary scans and HC PL APO 20x 0.70NA objective for image acquisition. For intestinal tissue cores arranged in a microarray we found that a 20X objective provides the optimal combination of high resolution and manageable file size. Other objectives can be used, depending on the samples and goals.

Filter cubes: care should be taken to ensure that the filter cubes adequately isolate fluorochromes without significant bleed-through between channels. The system described here is equipped with Chroma ET filter sets 49000, 49002, 49008, and 49009, e.g., for detection of Hoechst 33342, Alexafluor 488, Alexafluor 594, and Alexafluor 647, respectively. These filters are also suitable for imaging fluorochromes with similar excitation and emission wavelengths. The high-throughput Modified Magnetron (ET) sputter-coated line of filters was selected for durability and long life. The filters were mounted in cubes for Leica DM series microscopes and aligned by the

manufacturer to ensure perfect registration. This was validated between wavelengths using multi-fluorescent beads. To minimize movement of the filter turret and accelerate imaging, the filter cubes should be installed with the cube for the ubiquitous label, e.g., nuclear stain, in position 1 followed by the cube for the focus channel in position 2.

Camera: cameras with a wide-field-of-view and high sensitivity are preferred. CMOS cameras are ideal for this purpose. This system is equipped with an ORCA-Flash 4.0 LT+ CMOS camera (Hamamatsu). Although not essential, the system is also equipped with a second camera port that houses a Jenoptiks ProgRes C14-plus color CCD camera for transmitted light imaging, e.g., of hematoxylin and eosin-stained slides.

Software: this protocol and accompany journals are written for Metamorph 7 (Molecular Devices). Metamorph is used for post-acquisition processing and stitching, but open-source software, e.g., ImageJ/FIJI and ASHLAR can also be effective for stitching.^{4, 14} In either case, the resulting images are compatible with open-source software such as ImageJ/FIJI and GiMP. We use a commercial OMERO+ server with PathViewer (Glencoe Software) for image display. Open-source OMERO servers can also be used.¹⁵ If OMERO+ or OMERO are used, ASHLAR has the advantage that it creates the pyramidal TIFF files required by OMERO as part of the stitching process.

Vibration isolation: this is not required, but is preferred, depending on the location in which the microscope is housed. The system used here is mounted on a TMC antivibration table with air pressure-based dry damping.

EQUIPMENT SETUP.

Configuration of the microscope with the appropriate filter cube, camera(s) and motorized stage is pre-requisite to this protocol. Our setup has the minimum requirements with few added peripherals. Control of the automated filters and motorized stage must be integrated with the microscope control software. Metamorph meets these needs. The open source software μ Manager also has extensive hardware support that may be compatible with the system described.

Camera/stage xy alignment and field-of-view selection are critical. The camera must be aligned with the xy movements of the stage. This can be accomplished by tracking an identifiable object along the edge of the field-of-view and making incremental adjustments to the camera's rotation

until the object glides gently along the edge of the field-of-view with minimal drift. The camera's position, once adjusted, must be made secure. In addition, the field-of-view should not warp at the corners. Blurring in the corners must be eliminated by reducing the size of the effective field-of-view in the software. This is available in Metamorph via an option in the acquisition settings dialogue box. The Metamorph scanning module helps with this by staggering tiles slightly to reduce the impact of image defects at the corners. Selection of a planar objective, with minimal field curvature is important to maximizing the effective field-of-view. In this system, 80% of the camera's maximum field (13.312 mm^2) with the Leica HC PL APO 20x 0.70NA objective is an adequate compromise between gained speed of large fields of view and loss of resolution at tile corners. When stitching these images using ASHLAR¹⁴ we apply a light Gaussian filter to compensate for the distortion. Metamorph stitching handles these images effectively without additional corrections.

Stage z alignment is also critical. The first concern is flatness, the more level the stage is the more rapidly scanning will progress and the easier stitching will be. A perfectly flat stage would minimize the need for repeated focusing when collecting tiles, which would dramatically speed up scanning. By minimizing stage tilt, the focus point becomes increasingly consistent, minimizing the necessity of periodic focal checks and increasing their accuracy. The other benefit to a relatively flat stage is that stitching large ROIs that are not flat creates image distortion. We have found that a four-slide stage with a vertical change of 300 microns over the combined width of four 75mm x 25mm slides, or a tilt of 0.086-degrees, is within tolerance. The second concern is the ease of which the slides can be replaced in the same exact position, which is necessary for making multiple scans of the slides, such as for background subtraction and multiplexing. In our set up there is wiggle room between the slides, but pushing the slides towards two edges of the stage and securing them with slide holders eliminates this problem.

Metamorph requires some configuration to work with the provided journals. For Metamorph, it is essential that the ScanSlide module be installed and that the journals and acquisition files be copied onto your hard drive in the appropriate folders. ScanSlide receives xy coordinates for each ROI and collects a tiled image set according to the filter cubes, exposure times, and z offsets selected in the ScanSlide dialogue. As noted above, μ Manager has plugins that replicate some of these functions, including tiled scans. This may make it possible to translate the journals provided here for use in that open-source program. Similarly, the journals may be translatable for use in proprietary software provided by microscope manufacturers.

The Metamorph journal files can be stored anywhere, e.g. *MM/App/Journals/*. Regardless of location, once they are copied, some edits to these journals are required so that they correctly call each other and work with your equipment. Detailed instructions for this are written in text files that accompany the journals and the journals themselves are replete with comment fields explaining what each section accomplishes. Our goal is to provide flexibility and introduce the user to the mechanics of Metamorph journaling. The default directory structure for the journals is *C:/MM/App/TMA/*.

Acquisition parameters are set in three places. The settings that are likely to be regularly changed are adjusted on a per use basis and set in the ScanSlide application module. The settings that are used for the journal focusing routines are stored in *.AST* files located in the subdirectory *C:/MM/app/TMA/acquisitions/*. There are eight of these files in total, grouped in two sets of four, each of the four representing settings for one channel. They each define channel, exposure time, binning, field dimensions and other miscellaneous parameters. One set is used for tissue checks (e.g. *tissuecheckDAPI3XBINCENT.AST*) and the other set is used for autofocusing (e.g. *autofocusingDAPI1XBIN FULLCHIP.AST*). These setting can all be modified using the Acquire dialogue box in Metamorph and should be tailored to your camera's sensitivity.

Procedure:**Stage 1: creation of an overview image showing all tissue locations**

TIMING: approximately 15 minute set-up plus variable scan time (depending on area scanned and parameters selected; ~15 minutes per slide using settings and materials as described)

1. Slide preparation. Take care to avoid including air bubbles and debris in the mounting medium. For non-immersion objectives we generally use SlowFade Diamond or ProLong Glass, depending if we intend to rescan after tissue manipulation, e.g., other stains, or require archival storage, respectively. Just prior to imaging, ammonia-free cleaning solution, e.g., Sparkle, should be used to minimize focusing errors and image distortion caused by oils and fine debris. When multiplexing or scanning for background subtraction, securing the coverslip while cleaning by any reversible means, e.g., VALAP.¹⁶
2. Load slides onto the microscope stage. If the slides are going to be scanned twice or more, such as for background subtraction or multiplexing, push the slides tightly towards two edges of the holder, clamp them down at both ends and make a record of their positions. Anchoring is critical because any change in position will result in inaccurate ROI placement for additional scans.
CAUTION: If the slides are not replaced exactly when scanning is repeated, the multiplex scans may not resemble each other sufficiently to align images according to the reference channel.
3. Calibrate the *xy* control of the automated stage controller. One convenient and reproducible approach is to move the stage to the measurable limits of the linear encoders and setting this as (0,0).
4. Set the focus. Using a 5X objective, focus on a tissue near the center of the TMA or near the center of a multi-slide layout. Then, configure the software so that this is 0 in the *z* dimension. In Metamorph, this option is available under Device > Focus > Configure > Set Origin. Tight focus is not needed to create this map of the tissue layout, to be used in Stage 2, so a central focal point will suffice.
5. Load the scan settings. For this step, we use the built-in scan function in Metamorph called Scan Slide, which is selectable from the Applications drop down menu. If there is an existing settings file for collecting the 5X preliminary scan, load it now. *5X TMA Prelim Scan.scansetting* is provided as a sample settings file for this use.
6. In Scan Slide, select wavelength imaging and set the filter cube and exposure time. Set the exposure for the lowest interval that allows reliable tissue detection. We typically image nuclei stained with Hoechst 33342. There is no need to scan more than one channel for this overview image.

7. Perform a field-of-view calibration in ScanSlide. Select the objective to be used to make the preliminary scan; we find that the 5X objective works well at this step. This calibration procedure corrects for imperfect alignment between the automated stage movement and the orientation of the camera. An alignment correction of less than 1 degree is desirable. The calibration also calculates the pixel size, thus it is necessary to perform this calibration for each objective used in this procedure.
8. Define the region to scan in ScanSlide. Set the scan field by selecting the two opposing corner positions for the rectangular region. The region must include all of the tissues that will be scanned. We recommend extending the corners slightly beyond the anticipated area needed to avoid having to repeat this scan.

CAUTION: Avoid damage to the objective by ensuring that the slide clamps do not create an obstacle at the extreme edges of the region to be scanned.

9. Set the save folder. The save folder will contain all the preliminary scan images. For example, *C:\Data\user\YYMMDD*.
10. Save the scan settings. Once the 5X scan settings have been saved once, you can load them up before executing Stage 1 of the protocol and save adjustments made as needed.
11. Start the scan. As Metamorph scans the slides a showing the progress will open.
12. Observe the beginning of the scan to gauge whether exposure, tile alignment, and that the first column contains all the expected tissues. If needed, the scan can be aborted and repeated after correcting the errant settings.

Anticipated result: After the preliminary scan is completed, Metamorph provides an overview alignment of all the tiles showing the location of all the tissues to be scanned.

TROUBLESHOOTING: See Table 2.

Stage 2: selecting ROIs

TIMING: For four slides with 40 TMA positions plan for a 1 hour set up time plus variable scan time (overnight scanning of 4 slides is feasible with described materials and settings).

13. In Scan Slide, load the scan slide settings for the TMA scan. If these haven't been saved yet, work from the existing settings.
14. Change to the desired imaging objective, e.g. 20X, and calibrate the stage. Repeat the ScanSlide calibration procedure performed in Step 7 with the higher power objective. Although the angle calculated for the lower power objective should be unchanged, the process must be repeated to correct for pixel dimensions. Again, the camera should be well enough aligned that the calibration offset is not more than one degree.

CRITICAL STEP: If calibration is not performed correctly, the tiles will not align when stitched.

15. Define the ROIs. Use the rectangular region drawing tool to define each ROI (tissue) on the preliminary scan made in Stage 1. Consider how the slides are oriented and draw the first region in a logical position and continue drawing region in an orderly manner, e.g., top-down, left-right manner along the TMA. We recommend that tiny regions be placed at empty TMA positions to simplify orientation. Continue this procedure across all of the slides included in the overview scan. Follow a logical sequence so that the assigned ROI numbers can be interpreted later, e.g., left to right on the slide holder. A time-saving approach is to first quickly lay out regions of the same size on each tissue core and then return to adjust the size and position of each ROI to match each tissue. To save time during Stage 3, the region boxes should be tightly drawn around the tissues. The region shape defines the minimum area to be scanned, so some extra scanning around the perimeter will be collected.
16. Save an image of the regions superimposed on the preliminary scan to serve as a reference map by selecting Edit > Duplicate > As Displayed from the dropdown menu. It is helpful to save this in the same directory as the tiled images as a record of the numbering of each core. It is also useful to open this in a separate window, i.e., in a different program, so that it can be referred to during scanning.
17. Calibrate the position of the regions to match the automated stage movement. In general, changing objectives causes the regions placed on the overview 5X scan to be slightly displaced at 20X. This misalignment is uniform across all of the slides and is easily corrected.
 - a. In the preliminary overview scan (with all regions defined), select a ROI surrounding a core with an identifiable tissue shape.
 - b. Use the ScanSlide options to send the stage to the center of the ROI (it may also be helpful to move the stage to the corners of the ROI, which is also selectable in ScanSlide).
 - c. Using the 20X objective, view the imaged region on screen. It can help to look through the oculars or temporarily switch to the 5X objective for a wide field-of-view. If using the 20X or 5X objective, ensure that entire image is shown in the display (reduce the zoom magnification if needed).
 - d. Assess ROI location relative to tissue imaged. To correct misalignment, select Region > Move all Regions from the dropdown menu and use this tool to adjust ROI placement.
 - e. Repeat step 17b and again judge if the camera is capturing the intended part of the ROI. Repeat this process until you are satisfied that the ROI placement causes the stage to move such that the camera collects the correct image. At this point, the ROIs may not

appear to be centered on individual tissue cores when viewing the overview 5X scan. This doesn't matter as long as the stage is moved to the correct position for image capture.

- f. Double check your work with a different region.
 - g. If you looked through the oculars or used the 5X objective during alignment, be sure to reset the camera/ocular selection plunger or return to the 20X objective, respectively.
18. Save the regions. Select Regions > Save Regions from the dropdown menu. This is especially important if the slides will be rescanned because each scan must be the same dimensions. It is helpful to save this in the same directory as the tiled images.
- Anticipated Result: After completing this stage, all the regions of interest will have been defined and their locations saved for future reference.

Stage 3: setting imaging parameters and starting the automated scan

TIMING: Approximately 10 minutes

19. Determine the z range needed to find focus across all the ROIs of all slides to be imaged. Typically, a small variation in focal plane occurs as a result of imperfect planar alignment of the stage and camera. Hence, the focal points at the corners of the whole area being scanned are at the upper and lower limits.
- CRITICAL STEP: If the z range determined is too small the software will not be able to focus ROIs outside of that range. Alternatively, setting the z range too large will extend the time needed for focusing (and scanning).**
20. Set the z origin. Calculate the middle point of the focal limits determined in the previous step, then find tissue near the central position and determine if this estimate is approximately correct. With the active z position at the desired origin, select Device > Focus from the dropdown menu and click "Set origin" in the dialogue box.
- CRITICAL STEP: If the z -origin is not set, or set incorrectly, prior to initiating the scan, many of the ROIs will be out of focus.**
21. Change the scan settings to match your needs.
- a. In ScanSlide, select the number of required channels, their illumination settings (e.g., filter cubes), exposures, and any focusing offsets between channels. When scanning tissue labeled by multiple fluorochromes and a nuclear stain, often the focal point of the nuclei is displaced from that of the other fluors, which may also be displaced from each other. Each illumination setting will be used in the order presented in ScanSlide, from top to bottom. Each has the potential to include a z offset from the previous focal point. Take

advantage of this feature by determining the focal points in each channel and imaging them in a logical order. Ideally, this should be the same order in which the cubes are installed in the filter turret to minimize cube-changing time. For example, begin with the channel used for the ubiquitous label, e.g., the nuclear stain, in position one and the channel used for focusing in position 2.

- b. Carefully select the signal that will be used to achieve the initial focal point. We recommend that a signal that is always present at sites to be analyzed and has contrast to assist with focusing. When imaging tissues, a membrane stain, either of all cells or a cell population, e.g., E-cadherin for epithelial cells, a good choice. Signals that have multiple points of contrast over several microns of thickness, e.g., nuclei, may lead to variable focus across the stitched ROI image. Signals that are absent in some areas can also result in focus drift during the scan.
 - c. (optional) Set shading correction. This feature is useful if your illumination fades mildly at the corners of the image.
22. Set the save file path and name. Each set of scans should be in a separate own sub-folder within the parent folder for all the scans to be made of the TMA slides. For example, *C:\Data\user\YYMMDD\stain-used\c1/f*. to indicate the stain used, e.g., antibody, cycle number (c) and that this folder contains stained foreground (f) scans from which pre-stain background (b) scans will be subtracted. The file name should be descriptive and end with the number 1, assuming that the first ROI is ROI 1, e.g. *Scan 20x obj1*.
23. Save the scan settings in the same folder.
24. Run the script for the scan. From the drop down menu, select Journal > Run Journal and then select the master journal *TMA Scanning Master.JNL*.
25. As directed by the journal prompts, several user inputs are required:
 - a. Choose between two focusing routines, 1-point or 2-point, to determine the base focal position for each ROI.

The 1-point focusing routine begins at the center of the ROI and checks for tissue in the selected focusing channel. The tissue check procedure takes an image of the field in the user-selected channel. After scaling the image to remove the brightest spots (e.g. debris, aggregates of secondary antibody), the algorithm compares the upper limit of the scaled intensity range to a user-defined threshold, e.g., 2% for minimal peak luminance (the value is entered in a prompt below). If there is tissue, the microscope is focused. If no tissue is found, the stage moves to another position and repeats the process. This routine checks nine fields total (the center and 8 positions around the center; both movement

distance and pattern are explained with comments in the journals) before giving up and simply using the previous ROIs gross focus point.

The 2-point focusing routine follows the same search pattern as 1-point but compares focal point z values from two fields. If the two are not within 5 microns (explained by comments in the journals), a third focal point is collected. If either of the first two agree with the third (within 5 microns), that pair is averaged for the gross focal points. If very large ROIs with tissue that is not flat are being scanned, the 1-point routine should be used along with frequent deep refocusing (selected in Step 25.f.ii). For smaller ROIs, particularly small ROIs that have folded tissue, which can lead to incorrect autofocus, the second 2-point is preferable.

- b. Choose between two focusing routines to for fine focusing at each tile position: either focus at every tile position using the built-in ScanSlide focus routines or use the journal-programmed routine that checks focus only at user-defined intervals. The built-in focus routine for ScanSlide is limited to a single focusing step which is executed in every tile position, regardless of tissue presence. This routine can be triggered for a single wavelength or multiple wavelengths. The journal-programmed routine offers normal fine focusing and gross focusing in a user-selected channel, the focusing steps can be independently controlled by user-defined frequency, tissue is checked for in the same channel as the focusing before focusing occurs. The journal-programmed routine was developed to save scanning time by reducing the frequency of focusing, which we find does not change from field-to-field so often to require regular refocusing. It helps keep very large ROIs in focus by periodically adjusting the gross focus before making fine focus adjustments.
- c. Follow several prompts that ensure the necessary preparatory steps have been executed.
- d. Indicate if the appropriate Stage List Array is loaded. The Stage List Array file is created by utilizing Scan Slide to save all the xy coordinate positions of the currently drawn regions along with Scan Slide acquisition and calibration settings. This array file is used to execute the focusing and scanning routines progressively from ROI to ROI. For the first set of scans the array file needs to be constructed, however, if you are repeating a scan and the ROIs haven't changed, the Stage List Array file may be loaded by simply opening the previously-saved TIFF format file. Likewise, if repeated scans of the same slides are being executed, as in multiplex staining procedures, the Stage List Array saved during the initial run can be used.

- e. Input the z range. This is the maximum distance around $z = 0$ required for finding focus on any ROI in the TMA. This value was determined in Step 19. It is a good idea to include a buffer at both estimated focal limits, we typically add at least 10 microns.
- f. Input the focusing parameters to be used while imaging each ROI.
 - i. The gross focus range, which should be much less than the entire z -range, unless you are using these journals to scan a single large ROI. The step size between focus checks for this range 3 micron intervals; the variable that adjusts this can be found in *ROI focus.JNL* the position indicated within the comments.
 - ii. The gross focus threshold should be set to ensure that tissue is present before gross focusing is performed. The purpose of the threshold is to prevent focusing on a field with fluorescent debris. Focus on the center of an ROI and then find fluorescent debris in a field that does not contain tissue. Do not refocus on the debris. Run the *threshold finder.JNL* and follow the prompts. Use this journal as a tool to find the intensities of debris and real tissue, then set your threshold value somewhere between.

NOTE: exposures used for threshold checks are determined by an acquisitions setting file (.AST) which are all stored in the subfolder /acquisitions/ and may require alteration to match you camera's sensitivity.

- iii. The frequency of rechecking the gross focus, in terms of number of images/tiles taken before the gross focus is re-assessed. A fine focus check is automatically made after every gross focus check. Also see step v below.
- iv. The channel of interest for the gross focus routine. A nuclear stain is a fine choice, as it is bright and ubiquitous. The exposure used will be what was set for saving images.
- v. The fine focus range, usually just a few microns.
- vi. The step size between focal points in the fine focus range. Check your stage's movement accuracy before setting this parameter. It is best to determine what setting works best for specific samples beforehand by autofocusing using the Adjust Focus (not Find Focus) feature of Metamorph, which is located in the Devices > Autofocus via software dropdown menu.
- vii. The fine focus threshold is determined in the same manner as the gross focus threshold (see step ii). It is critical to adjust this appropriately when the channel used for fine focusing is not the same as the channel used for gross focusing. If

the same channel is used for both gross and fine focusing enter the same value as entered in step ii.

- viii. The frequency of fine focus checks, in terms of number of images/tiles taken before the gross focus is re-assessed. To determine how often this is needed, in advance of scanning, focus on one field-of-view at one edge of a tissue and, then, move in one direction by full-field steps. A value equal to or smaller (more frequent focusing) than the number of fields interval over which focus is lost should be entered. Ideally, the frequency of rechecking the gross focus should be a multiple of this value.
- ix. The channel to use for fine focusing. If the label, e.g., protein, of interest has a more restricted *z* focal plane than the nuclei, this should be selected for fine focusing. This may be different than the channel used for gross focusing at the start of each ROI, i.e., 1-point or 2-point routines.

TROUBLESHOOTING: See Table 2.

- 26. The journal will now run without any further user input. First it will move to the coordinate positions (opposite corners) of each ROI to build a Stage List Array file. The stage will then move to the center of the first ROI (ROI 1) and the gross focal point for that ROI will be determined using the 1-point or 2-point routine, as selected above. The journal then calls ScanSlide to image the ROI from upper left to lower right in a snake-like pattern. When tissue is found, the fine focus will be adjusted before taking an image. Gross- and fine-focusing will be performed periodically, according to the parameters chosen in Step 25.
- 27. After ROI 1 is fully imaged, the stage moves to ROI 2 and repeats the process. Watching acquisition of the initial field to ensure that the process begins correctly is recommended. Anticipated result: After scanning all the ROIs the save directory will be populated with all the associated raw data files (TIFF format) and a *.scan* file for each ROI. The latter files contain all the organizational information about the data files for the ROI. A stage list array file is also generated in the parent folder to the image folder with the name of the save folder, e.g., *C:/Data/user/YMMDD/stain-used/c1/fStage list array.tiff*.

Stage 4: review the scans for imaging errors

TIMING: 1/2 minute assessment per ROI plus scan time.

- 28. Copy the stage list array file to the image folder. Find the stage list array file in the parent folder to the image saving folder that was selected in Scan Slide. This file is used to guide the

Metamorph journals for stitching together all the ROIs in a single command. If this file is overwritten or lost, the process of stitching by Metamorph must be done individually for each ROI, which is very time consuming.

29. Examine each scan to determine if there were focusing errors or if any ROIs were scanned incompletely. Using ScanSlide, open the *.scan* file for each ROI individually. This will create a low resolution image of the ROI that can be used for assessing region imaged and gross focus errors.

TROUBLESHOOTING: See Table 2.

30. Rescan ROIs as needed.
 - a. In Scan Slide, load the Preliminary Scan *.scan* file, found in the save directory specified during Stage 1.
 - b. Load the regions file made at Step 18. From the drop down menu select Regions > Load regions.
 - c. Make a sub-folder in the original image folder, e.g., *C:\Data\user\YYMMDD\stain-used\c1\f\rescans* and set the save path to this folder.
 - d. Open the settings *.scansettings* file within ScanSlide.
 - e. For all ROIs that were scanned inadequately, make appropriate adjustments to those regions and save the changes as a new regions file. Take care to save under a different name so that the original regions file is not overwritten. The new regions file will be used if the slides are to be rescanned. Take care to save under a different name so that the original regions map is not overwritten.
 - f. Click on the first region that needs to be rescanned.
 - g. Set the file name to match that of the original files, e.g. *Scan 20x obj3* if the ROI 3 needs to be rescanned.
 - h. Move the stage to the ROI being rescanned and focus manually.
 - i. Activate ScanSlide.
 - j. Repeat 30f through 30i for each region the needs to be rescanned. During the journal prompts, indicate that there is no existing stage list array.

Alternate approach: after Step 30e, delete the ROIs that were scanned correctly and re-save the regions file. The protocol can then be re-initiated at Step 24. However, this will name the ROI image files sequentially, beginning with 1, e.g., *Scan 20x obj1*, even if the first ROI rescanned was originally ROI 3, in the new save folder. It will therefore be critical to record this new numbering strategy. TIFF file names must then be corrected to the original ROI number. Copy the TIFF files to the original acquisition folder. **CAUTION: This will**

overwrite the original TIFF files. Be careful to retain the original *.scan* file, as this includes the image file names used during acquisition and the image files have been renamed.

31. After the scan is complete, copy the new stage list array into the new save folder.

Anticipated Result: All ROIs should be scanned correctly. This should be verified as in Steps 29 and 30.

Stage 5, (optional) rescanning the TMA slides for background subtraction and antibody multiplexing

TIMING: ~15 minutes plus scan time

32. Clean the slides carefully, as in Step 1.

33. Load the slides back onto the stage, as in Step 2.

CRITICAL: If the slides are not loaded exactly as they were before, the scans will not align correctly. Some error is tolerated, but it should be minimized. Slide holders that hold each slide tightly in a precise position are recommended.

34. In ScanSlide, load the saved scan settings from the previous scan. The scan settings file contains all the exposure information, which should be the same in both sets of scans.

35. In Scan Slide, load the previously made scan *TMA_Prelim Scan.scan*. This will bring up the low-resolution scan of all the slides. If the previously generated Stage List Array file was saved, it can also be opened from the drop down menu by selecting File > Open.

36. Load the previously made regions file. From the drop down menu, select Regions > Load regions. The map of the regions previously scanned will now appear over the image.

37. Confirm that that the slides have been positioned exactly as they were during the initial scans.

- a. Select a ROI surrounding a core with an identifiable tissue shape.
- b. Use the ScanSlide options to send the stage to the center of the ROI (it may also be helpful to move the stage to the corners of the ROI, which is also selectable in ScanSlide).
- c. Using the 20X objective, view the imaged region on screen. It can help to look through the oculars or temporarily switch to the 5X objective for a wide field-of-view. If using the 20X or 5X objective, ensure that entire image is shown in the display (reduce the zoom magnification if needed).
- d. Assess ROI location relative to tissue imaged. If it is not correct, reposition the slides to match the original placement. Do not move the ROIs (as in Step 17).

- e. Double check positioning with a different ROI. If it is not correct, repeat Step 37d until all ROIs align (checking a few on each slide, e.g., the center and one or two corners, should be sufficient). Do not proceed to Step 38 until slides are correctly positioned.
- f. If you looked through the oculars or used the 5X objective during alignment, be sure to reset the camera/ocular selection plunger or return to the 20X objective, respectively.

Set the file path for saving the images. We recommend that you save each scanning cycle of the TMA slides in a different sub-folder within a parent folder containing the preliminary scan. For example, *C:\Data\user\YYMMDD\stain-used\c1\f*, as described in Step 22.

- 38. Run the journal as in Step 25. Ensure the reference channel is set to the ubiquitous stain, e.g., nuclei, that will be used to align the repeated scans. If the reference channel is out of focus between scan cycles it may not be possible for ASHLAR to align the scans for subtraction.
 - 39. Copy the stage list array to the image folder.
 - 40. Review the scans as described in Stage 4 to identify unacceptable, e.g., poorly-focused scans. Do not make any changes to the region positions; alignment requires that all ROI locations must be precisely the same as the previous scans. Delete all regions with acceptable images so that only ROIs over regions that were not imaged well remain. Take care to save this ROI set with different name, e.g., the *rescan* folder, so that the original regions map is not overwritten.
- Anticipated results: A new set of scan files will be created in the new sub-folder. The scanning set up should be easy and the results should be as expected.

Stage 6, assembling the image tiles into stitched images for analysis

TIMING: ~15 minutes plus data transfer and processing time

- 41. Copy the data to a workstation or other computer. Stitching is processor intensive, but most PCs with reasonable power can stitch 40-50 ROIs with the Metamorph stitching routine in about 1.5 hours.
- 42. Use Metamorph or ASHLAR to stitch the images.
 - a. Stitching with Metamorph:
 - i. Open the stage list array file of the scan. The stage list array file must be in the same folder as the images it pertains to.
 - ii. From the dropdown menu, select Journal > Loop > Loop for all Images
 - iii. In the dialogue box select the journal *Review TMA scans.JNL* and initiate the procedure. This routine will create a subfolder *Stitched Images* in the folder where the images are saved and saves the stitched images to the folder. Metadata from the scan is included in the stitched image, e.g., stage position, exposure

time, objective, channel, camera bit depth, stage hardware settings, and time of creation.

- iv. (optional) If you want to create multi-channel image stacks of all the regions, then select Journal > Loop > Loop for all Images in Directory. Select the directory in which the images are saved and run the journal *collect multiple wavelengths into stacks.JNL*.
- v. (optional) Most virtual microscope applications, including OMERO, require a pyramidal image file. This can be created from the multi-wavelength image stack using the Python-based program *Stitcher* (Glencoe Software) with the *pyramid* option.

b. Stitching with ASHLAR:

- i. ASHLAR is open source software under continual refinement and is available for download from GitHub. ASHLAR runs in Python and has no graphical user interface. Some fundamental knowledge of Python is therefore required for its use. Unlike Metamorph, ASHLAR can align several staining cycles of the same tissue based on a reference channel from each cycle. The data must be described by a *.scan* file and the reference channels must all be the same number (the default reference channel is 0). This feature makes ASHLAR suitable for multiplexing and background subtraction experiments. The output is a series of stitched TIFF files ready for analysis.
- ii. Determine the necessary parameters required to stitch the tiled images. Tiled images collected in Metamorph are listed in the reverse order from the ASHLAR default. Command line options (*--flip-x* and *--flip-y*) have been integrated into ASHLAR to correct for this. If there are sensor noise issues in the corners of tiles, this can be offset by applying a Gaussian filter (*--filter-sigma SIGMA*). It is recommended that you begin with a filter of 1 and increase as needed. When run from the home directory of a scanning project with multiple subdirectories containing image files, the following command line would stitch two cycles of staining in the same way, aligning by the default channel (0) and saving them to the subdirectory *stitched*:

```
ashlar --flip-x --flip-y --filter-sigma 1 /c1/f/Scan 20x obj1.scan /c1/b/Scan 20x obj1.scan /c2/f/Scan 20x obj1.scan /c2/b/Scan 20x obj1.scan --o /stitched/
```
- iii. Write a Python script to run ASHLAR on all the scans you need to stitch. The process of stitching is time intensive, especially if multiple imaging cycles are

involved. To save user interface time, write a looping script to execute ASHLAR on each *.scan* file, each of which represents an ROI. Some basic knowledge of Python programming is required for this action.

- iv. (optional) ASHLAR has built in functionality to output pyramidal TIFF image files.

Timing:

Most of the time required is the automated scanning. This will increase linearly with the number and size of ROI being scanned and the imaging parameters. Other variables that bear considerable consideration include size of the camera capture area, the speed of the motorized filter cube turret, and to a lesser extent the speed of the automated stage. The time of filter cube exchange can be eliminated through use of a wavelength-selectable light engine, e.g., LEDs or lasers, and multi-bandpass dichroic and emission filter. Our experience, however, is that this increases the tissue autofluorescence signal captured.

The used of high NA objectives with limited depth of field requires many focusing steps during scanning. Setting this optimally will maximize success while minimizing scan time. Before each ROI is scanned, a gross ROI focal point is determined. We provide two options for how the focal point is found. The more rapid option takes the strategy of beginning at the center of the ROI, checking for tissue and if tissue is present the focus for the ROI is taken there. If insufficient tissue is found then the camera moves to neighboring fields-of-view until sufficient tissue is found and the gross focal point for the ROI is taken from there. The more conservative option is slower at this step, which is only done once for each ROI. This approach ensures that the focal point for the ROI is correct by searching for tissue and focusing. The routine is, however, repeated until two focal points are found that agree within 5 microns (this tolerance is adjustable by modifying the journal). The average of these is used as the gross focal point for the ROI. This helps to avoid misfocusing of an entire ROI, e.g., due to bright debris at the center of the ROI in a different focal plane than the tissue. Regardless of which routine is used, the size of the focal range being checked at each ROI, which is defined by the user and determined at Steps 18 and 19, will also affect the scan time. The step size, set in the journal, is 3 microns.

Regular focusing on each tile also affects the timing of the TMA scan. The provided journals allow for two options on how this is done. The first is to use the built-in focusing routine of Scan Slide, which will check the focus at every tile position and can be programmed to focus in one or all wavelengths. Alternatively, the focusing step can be made at a user-defined interval instead of on every tile. The number of focal planes checked to find the fine focus is user definable by step number and distance between steps. Intermittently during ROI scan a deep focus check can also be made. The frequency and the size of the deep focus check is user definable. As above, the default step size is 3 microns.

The increased data collection rate with these incremental changes is substantial and can have a vital impact on both experimental design and throughput. The rate-limiting step of multiplexing, for example,

is scan time. The more data that can be collected during overnight scans, the better, as stripping and reapplying antibodies can usually always be performed during the day. Scanned at 20X magnification, a 40 tissue TMA stained in four channels uses about 40 GB of data and takes 3 to 5 hours, depending on the camera's field-of-view and focusing routines (Table 1). When operating efficiently, it may be feasible to perform a complete multiplex cycle on one slide in 24 hours, depending on staining and stripping conditions. If longer antibody incubations are required, a complete cycle on 3 or 4 slides can be every 48 hours.

Camera Chip Size	ROI focusing routine	Scan Time (Gb / h)
1392 x 1040 pixels; 6.45 μm^2 pixels	Single position ROI gross focusing and 15 step focus on every tile	~3
1638 x 1638 pixels; 6.5 μm^2 pixels	Single position ROI gross focusing and 15 step focus on every tile	~6
1638 x 1638 pixels; 6.5 μm^2 pixels	Single position ROI gross focusing and 15 step focus on 1/5 tiles, 10 step gross focus on 1/15 tiles	~8
1638 x 1638 pixels; 6.5 μm^2 pixels	2-point ROI gross focusing and 15 step focus on 1/5 tiles, 10 step gross focus on 1/15 tiles	~7.5

Table 1. Scan speed with different scanning conditions variables

Troubleshooting:

Step	Problem	Possible Reason	Solution
12, 28	The tiles are not aligned or are the incorrect size	Objective magnification was set incorrectly in ScanSlide	Repeat the ScanSlide calibration and re-start the protocol at Step 14 (Step 7)
12	The area scanned does not include all the tissues	The two corner limits of the scanning area were set incorrectly	Repeat Step 8 using corners limits that are further apart and rescan.
25	The journal cannot find a referenced journal and returns errors	(i) The journal files are not in the correct folders (ii) The journal files have been renamed	(i) Take care to correctly modify the journals during set up so that each calls the correct folder for the referenced journal (ii) Do not rename the files unless the journals are modified to reflect the new names
28	All images from one area of the slide(s) are out of focus	(i) Tissues are outside of the of the focus range (ii) The focal range was centered around the incorrect origin (iii) The stage is not parallel to the camera	(i) Select a larger z -range for gross focus is too small; If the overview scan and ROIs are acceptable and the slide(s) has not been moved, repeat the protocol beginning at Step 19 (ii) The z origin was set incorrectly, repeat the protocol beginning at Step 20 (iii) Correct z alignment of the stage as described in the Equipment Setup section
28	The wrong areas were scanned	The ROI boxes were not correctly aligned	Reset the ROI positions. If the overview scan and ROIs are acceptable and the slide(s) has not been moved, re-start the protocol beginning at Step 17

Anticipated Results:

When scanning is successful the focus will be clear in each and the seams between tiles are nearly imperceptible (Fig. 2A). When scanning is only moderately successful a Metamorph-stitched image will be skewed (Fig. 2B) and if it is completely unsuccessful it will not be possible to stitch together the tiles. Once good parameters have been found for scanning TMAs, the only major challenge to maintained success is keeping the camera aligned. Data management is a separate challenge.

Using this described procedure we have scanned numerous TMA slides of human and mouse intestinal tissues with 4-channel immunofluorescence. Over 5,000 scanned images have been compiled into the publically-accessible Atlas of Intestinal Transport (<https://www.jrturnerlab.com/atlasofintestinaltransport>). Each image was saved in pyramidal TIFF format, allowing the OMERO+ Pathviewer server to act as a virtual microscope, replete with threshold adjustments for each channel.

Figure Legends:

Figure 1. Placing ROI boundaries. *A.* Overview scan of two slides, each with a 50 core microarray. ROIs have been positioned over each core. Note that a small ROI is placed over positions without a core in order to maintain consistent image file numbering. *B.* Magnified view of the boxed area in A.

Figure 2. Representative data. *A.* Four color overlay of imaged tissue at low power (i) and an enlarged view of the boxed area (ii). Additional data is presented online (<https://www.jrtturnerlab.com/atlasofintestinaltransport>). *B.* Single-channel stitched image of nuclei collected after correct calibration of ScanSlide (i). When this (green) is combined with a second image (red) acquired with improperly calibrated ScanSlide (ii), a few areas align (left box, enlarged in iii) but most do not (right box, enlarged in iv).

References:

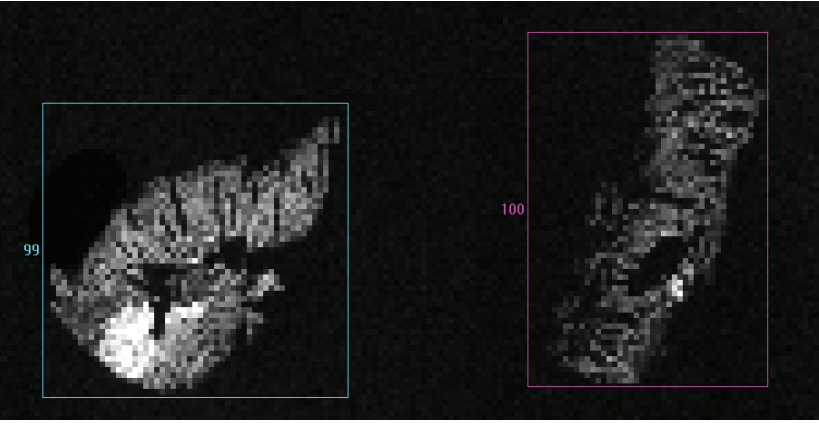
1. Dhir R. Tissue microarrays: an overview. *Methods Mol Biol* 2008; 441:91-103.
2. Robertson D, Savage K, Reis-Filho JS, et al. Multiple immunofluorescence labelling of formalin-fixed paraffin-embedded (FFPE) tissue. *BMC Cell Biol* 2008; 9:13.
3. Abramoff MD, Magelhaes PJ, Ram SJ. Image processing with ImageJ. *Biophotonics International* 2004; 11:36-42.
4. Schindelin J, Arganda-Carreras I, Frise E, et al. Fiji: an open-source platform for biological-image analysis. *Nat Methods* 2012; 9:676-82.
5. Schneider CA, Rasband WS, Eliceiri KW. NIH Image to ImageJ: 25 years of image analysis. *Nat Methods* 2012; 9:671-5.
6. Edelstein AD, Tsuchida MA, Amodaj N, et al. Advanced methods of microscope control using muManager software. *J Biol Methods* 2014; 1.
7. Bolognesi MM, Manzoni M, Scalia CR, et al. Multiplex Staining by Sequential Immunostaining and Antibody Removal on Routine Tissue Sections. *J Histochem Cytochem* 2017; 65:431-444.
8. Gerdes MJ, Sevinsky CJ, Sood A, et al. Highly multiplexed single-cell analysis of formalin-fixed, paraffin-embedded cancer tissue. *Proc Natl Acad Sci U S A* 2013; 110:11982-7.
9. Lin JR, Fallahi-Sichani M, Chen JY, et al. Cyclic Immunofluorescence (CycIF), A Highly Multiplexed Method for Single-cell Imaging. *Curr Protoc Chem Biol* 2016; 8:251-264.
10. Lin JR, Izar B, Wang S, et al. Highly multiplexed immunofluorescence imaging of human tissues and tumors using t-CyCIF and conventional optical microscopes. *Elife* 2018; 7.
11. McKinley ET, Sui Y, Al-Kofahi Y, et al. Optimized multiplex immunofluorescence single-cell analysis reveals tuft cell heterogeneity. *JCI Insight* 2017; 2.
12. Li W, Germain RN, Gerner MY. Multiplex, quantitative cellular analysis in large tissue volumes with clearing-enhanced 3D microscopy (Ce3D). *Proc Natl Acad Sci U S A* 2017; 114:E7321-E7330.
13. Preibisch S, Saalfeld S, Tomancak P. Globally optimal stitching of tiled 3D microscopic image acquisitions. *Bioinformatics* 2009; 25:1463-5.
14. Muhlich J, Russell DP, Chen Y-A. Alignment by Simultaneous Harmonization of Layer/Adjacency Registration. *GitHub* 2017; github.com/sorgerlab/ashlar.
15. Allan C, Burel JM, Moore J, et al. OMERO: flexible, model-driven data management for experimental biology. *Nat Methods* 2012; 9:245-53.
16. Chazotte B. Mounting live cells onto microscope slides. *Cold Spring Harb Protoc* 2011; 2011:prot5554.

Nilsen et al. Fig 1 .

A



B



Nilsen et al. Fig 2.

